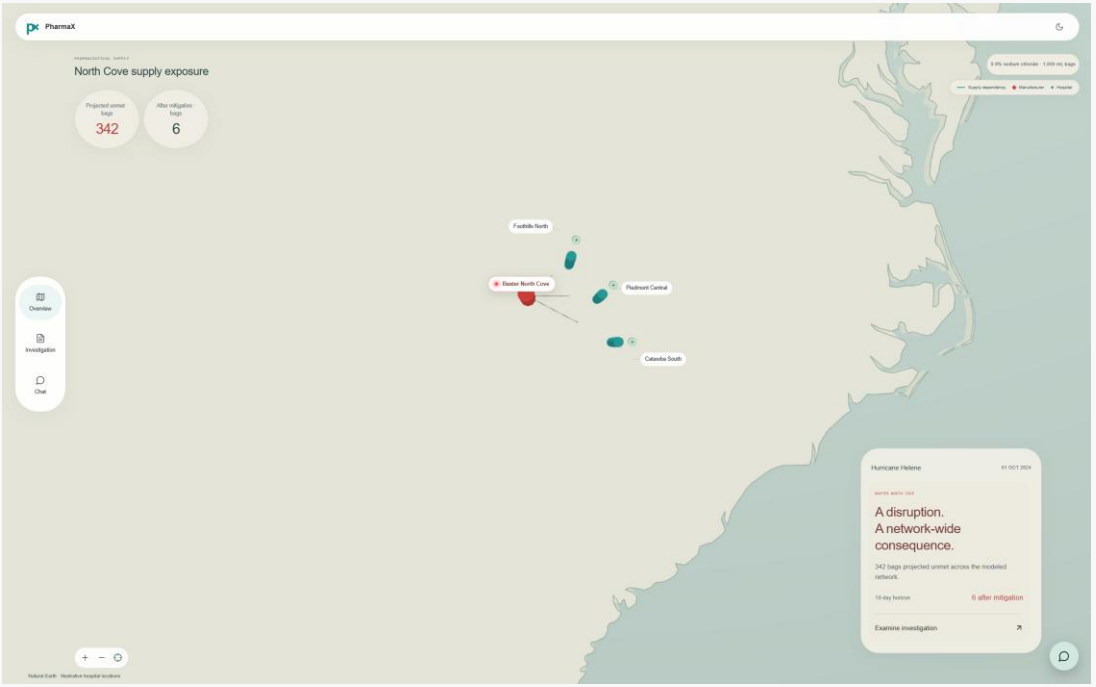
PharmaX

Spot medicine shortages early. Get essential supplies where they are needed.

Reasoning Architecture



For pharmacy supply teams: connect manufacturer notices, shipment claims and hospital inventory to predict shortages and recommend reviewed transfers.

When supply breaks, hospitals run short.

A factory disruption can leave one hospital short while another has spare stock. Teams need to find the gaps and evaluate safe transfers before supplies run out.

1 Supply disrupted

Manufacturing slows or stops. Fewer saline bags enter the network.

2 Deliveries fall short

Orders may be delayed or partially filled.
Available supply becomes uncertain.

3 Hospitals face shortages

Some hospitals risk running out; others may have transferable stock.

PharmaX maps the disruption, predicts unmet demand and identifies feasible transfers that protect both receiving and donating hospitals.

Solution

A single investigation links evidence to a shortage forecast and an actionable review draft.

01 Connect records → **02** Match medicines → **03** Check claims → **04** Predict shortages → **05** Propose transfers

342 → **6** modeled unmet saline bags after evaluated transfers

Scenario forecast · three synthetic hospitals · human pharmacy review required

Agent architecture

12 coordinated roles share evidence, challenge proposals and validate what can safely be recommended.



Evidence → assessment → critique → revised/blocked decision. Recalled failures constrain proposals; only supported findings reach the coordinator.

12 roles executed · 11 findings validated · Hospital finding rejected and excluded downstream

Technical architecture

From data retrieval to constrained decisions: one traced investigation pipeline.

01 SOURCE DATA	Zetaris	Join manufacturer allocation notices with hospital inventory and receipt records; normalize product codes.
02 AGENT SYSTEM	Nemotron	Coordinator and 11 specialists use Nightshift model calls and tools to examine evidence and draft decisions.
03 MEMORY + GUARDS	H-MEM	Recall rejected candidates; validate evidence, recalculate shortages and require pharmacy review.

Python orchestration + typed API → Zetaris source queries → Nemotron via Meterless Nightshift → FEFO forecast + review

Why the integrations matter

The data, model and memory layers constrain the same operational decision.

1	Zetaris	Cross-system product matching	Manufacturer item → NDC → local item code
2	Evidence verification	Shipping claim versus hospital receipt	800 shipped / 500 received → 300 unconfirmed
3	Memory limits proposals	Previously rejected options stay flagged	5 recalled records constrain subsequent review
4	Feasibility checks	Only reviewable options are proposed	Reject options that create donor shortages or assume unconfirmed supply

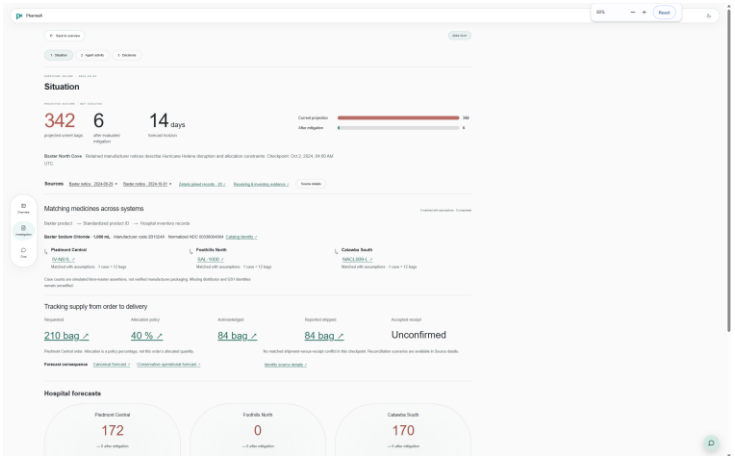
Distinctive: validated claims + inter-agent constraints, not 12 independent summaries

One case, three interfaces.

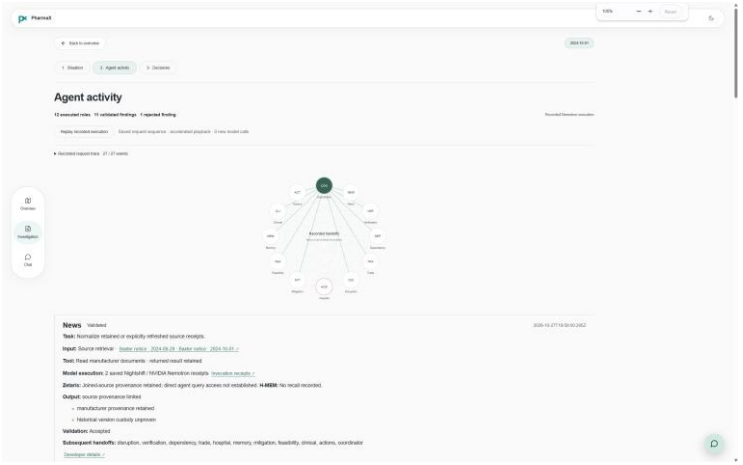
Judges can reproduce the narrative from overview to evidence to action.



01 Overview



02 Investigation



03 Agent execution

Dedicated sourced chat and review drawers complete the operator workflow.

Built from operational feedback

Industry advisor Kulathusarma highlighted the hardest deployment problem: linking records across organizations and verifying what was actually delivered. Feedback shared by Robbie Harris.

01	Connect systems	Join manufacturer availability and hospital demand records.
02	Check claims	Distinguish reported shipping from hospital-confirmed receipt.
03	Match medicine IDs	Match manufacturer product codes to hospital item codes, quantities and lots.
04	Review decisions	Test transfer feasibility and require pharmacy review before any action.

What we built in response: joined data sources, product matching, shipment-versus-receipt checks and feasibility-gated transfer drafts.

Future development

Next steps for turning a validated prototype into an operational tool.

01	Authorized hospital connectors	Connect real inventory, purchasing and receiving systems with permission.
02	Continuous monitoring	Refresh evidence and forecasts when supply or demand changes.
03	Stronger operational validation	Validate real approval roles, transfer ownership and confirmed delivery events.

Current scope: public Baxter documents, synthetic hospital data and simulated transfer proposals.